Name: Tharunya BS Register Number:250701786 Experiment name: Linear Regression

```
import numpy as np
import pandas as pd
df=pd.read_csv('Salary_data.csv')
df
```

	YearsExperience	Salary
0	1.1	39343
1	1.3	46205
2	1.5	37731
3	2.0	43525
4	2.2	39891
5	2.9	56642
6	3.0	60150
7	3.2	54445
8	3.2	64445
9	3.7	57189
10	3.9	63218
11	4.0	55794
12	4.0	56957
13	4.1	57081
14	4.5	61111
15	4.9	67938
16	5.1	66029
17	5.3	83088
18	5.9	81363
19	6.0	93940
20	6.8	91738
21	7.1	98273
22	7.9	101302
23	8.2	113812
24	8.7	109431
25	9.0	105582
26	9.5	116969
27	9.6	112635
28	10.3	122391
29	10.5	121872

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   YearsExperience  30 non-null     float64
1   Salary           30 non-null     int64
```

```
dtypes: float64(1), int64(1) memory
usage: 612.0 bytes
```

```
df.dropna(inplace=True) df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29 Data
columns (total 2 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
0   YearsExperience  30 non-null     float64
1   Salary           30 non-null     int64
dtypes: float64(1), int64(1) memory usage:
612.0 bytes df.describe()
       YearsExperience         Salary
count        30.000000      30.000000
mean          5.313333   76003.000000
std           2.837888   27414.429785
min           1.100000   37731.000000
25%           3.200000   56720.750000
50%           4.700000   65237.000000
75%           7.700000  100544.750000
max          10.500000  122391.000000
features=df.iloc[:,[0]].values label=df.iloc[:,[1]].values

from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(features,label,test_siz
e=0.2,random_state=42)

from sklearn.linear_model import LinearRegression
model=LinearRegression()
model.fit(x_train,y_train)
LinearRegression()
model.score(x_train,y_train)
0.9645401573418146
model.score(x_test,y_test)
0.9024461774180497
model.coef_
array([[9423.81532303]])
model.intercept_
array([25321.58301178])
```

```
import pickle pickle.dump(model,open('SalaryPred.model','wb'))
model=pickle.load(open('SalaryPred.model','rb'))

yr_of_exp=float(input("Enter Years of Experience: "))
yr_of_exp_NP=np.array([[yr_of_exp]])
Salary=model.predict(yr_of_exp_NP)
Enter Years of Experience:  44
print("Estimated Salary for {} years of experience is {}: "
.format(yr_of_exp,Salary))
Estimated Salary for 44.0 years of experience is [[439969.45722514]]:
```